Batuhan Altundas

Robotics | Machine Learning | Multi-Agent Coordination

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Research Summary

Computer Science Ph.D. student in the CORE Robotics Lab at Georgia Tech, advised by Prof. Matthew Gombolay. Develops graph neural networks and reinforcement learning methods for multi-agent task allocation, scheduling, and motion planning, with an emphasis on scalable, adaptive, and interpretable coordination for robot and human-robot teams. First author on papers at NeurIPS and IROS and a literature review on human-robot teaming.

Education

Georgia Institute of Technology
Ph.D., Computer Science; expected May 2028, Atlanta, GA
Aug 2023 - Present
Advisor: Prof. Matthew Gombolay | CORE Robotics Lab, School of Interactive Computing

Georgia Institute of Technology
M.S., Computer Science; Computational Perception and Robotics, Atlanta, GA
Aug 2020 - May 2022

University of Michigan
B.S.E., Computer Science and Mechanical Engineering (dual major), Ann Arbor, MI
Aug 2014 - Dec 2019

Research Experience

Graduate Research Assistant
Georgia Institute of Technology, CORE Robotics Lab, Atlanta, GA
Aug 2023 - Present

- Develop graph neural networks, heterogeneous graph transformers, and recurrent reinforcement learning policies for multi-agent allocation, scheduling, and path planning.
- Developing a journal extension of **HybridNet**, focused on modeling stochastic human task performance and learning across repeated rounds for human-robot team coordination.
- Developed **TARGETNET**, a residual heterogeneous graph transformer that jointly assigns tasks and generates schedules while accounting for agent capabilities, travel times, and temporal constraints (**NeurIPS 2025**).
- Evaluated TARGETNET on problems with up to **40 agents and 200 tasks**, generalizing from smaller training instances and achieving **20-250x speedups** over prior graph-based methods across benchmark scales.
- Derived interpretable decision tree representations of learned task allocation and scheduling policies.

Research Scientist I
Georgia Institute of Technology, Atlanta, GA
Jul 2022 - Aug 2023

- Led graduate researchers in multi-agent task allocation, scheduling, and motion planning.
- Developed simultaneous scheduling methods using a single forward pass for joint assignment and ordering, replacing task-by-task model evaluations.

Research Experience (Continued)

Research Assistant
University of Michigan, Ann Arbor, MI
Jun 2018 - Sep 2018

- Analyzed, modeled, and prototyped a robotic joint using a gearless clutch system for legged robots.

Industry Experience

Data Science Intern
Halliburton, Houston, TX
Jun 2021 - Aug 2021

- Applied unsupervised learning and dimensionality reduction to industrial time-series data in Azure Databricks.
- Trained variational autoencoders and autoregressive neural networks for feature prediction and anomaly characterization.
- Presented modeling approaches and technical findings to nontechnical stakeholders.

Teaching Experience

Graduate Teaching Assistant
Georgia Institute of Technology, Atlanta, GA
Aug 2021 - May 2022

- Supported instruction in **CS 6476: Computer Vision** for the Online Master of Science in Computer Science (OMSCS) program.

Conference Publications

Heterogeneous Graph Transformers for Simultaneous Mobile Multi-Robot Task Allocation and Scheduling under Temporal Constraints
Batuhan Altundas, Shengkang Chen, Shivika Singh, Shivangi Deo, Minwoo Cho, Matthew Gombolay. *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.

Learning Coordination Policies over Heterogeneous Graphs for Human-Robot Teams via Recurrent Neural Schedule Propagation
Batuhan Altundas, Zheyuan Wang, Joshua Bishop, Matthew Gombolay.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022.

Literature Reviews

Human-Robot Teaming: Grand Challenges
Manisha Natarajan, Esmaeil Seraj, **Batuhan Altundas**, Rohan Paleja, Sean Ye, Letian Chen, Reed Jensen, Kimberlee Chestnut Chang, Matthew Gombolay. *Current Robotics Reports*, 2023.

Workshop Papers

Improvement of Optimization using Learning Based Models in Mixed Integer Linear Programming Tasks
Xiaoke Wang, **Batuhan Altundas**, Zhaoxin Li, Aaron Zhao, Matthew Gombolay.
arXiv preprint arXiv:2506.06291, 2025.

Towards Learning Scalable Agile Dynamic Motion Planning for Robosoccer Teams with Policy Optimization
Brandon Ho, **Batuhan Altundas**, Matthew Gombolay.
RoboLetics: Workshop on Robot Learning in Athletics at CoRL, 2023.

Other Publications and Preprints

Interactive Knowledge Distillation with Adaptive Teachers in Cooperative Multi-Agent Reinforcement Learning
Minwoo Cho, **Batuhan Altundas**, Matthew Craig Gombolay.
Robotics: Science and Systems (RSS), 2026.

Towards Automated Semantic Interpretability in Reinforcement Learning via Vision-Language Models
Zhaoxin Li, Zhang Xi-Jia, **Batuhan Altundas**, Letian Chen, Rohan Paleja, Matthew Gombolay. *arXiv preprint arXiv:2503.16724*, 2025.

Patents

Batuhan Altundas, Zheyuan Wang, Matthew Gombolay. Learning Coordination Policies over Heterogeneous Graphs for Human-Robot Teams via Recurrent Neural Schedule Propagation. U.S. patent application No. 63/592,374, 2023.

Research Projects

LLM-Based Multi-Agent Coordination
Python, LLaMA, GPT, reinforcement learning

- Built an adversarial tabletop-game environment supporting reinforcement learning policies and natural-language commands.
- Fine-tune foundation models and evaluate their performance against reinforcement learning agents in multi-agent settings.

Behavior Cloning for Fetch Robot Manipulation
Python, OpenAI Gym, imitation learning, learning from demonstration

- Developed a custom simulation environment for Fetch robot pick-and-place tasks with execution noise.
- Evaluated noise-free pretraining for behavior cloning before training policies in noisy environments.

M-Bot Autonomous Navigation
Python, computer vision, SLAM, Raspberry Pi

- Integrated line detection and computer vision with localized SLAM to model indoor corridors and correct robot heading.
- Developed vision-based intersection detection and integrated path finding for semi-autonomous indoor navigation.

Professional Service

Reviewer: RA-L; ICRA 2027/2026/2024; ICLR 2026; NeurIPS 2025/2024; AAMAS 2025; IROS 2024; MRS 2023; CoRL 2022.

Technical Skills

Programming: Python, C++, SQL, MATLAB

ML frameworks: PyTorch, TensorFlow, Deep Graph Library (DGL), Pyro

Machine learning: Deep learning, reinforcement learning (RL), graph neural networks (GNNs), transformers, imitation learning, behavior cloning, large language models (LLMs)

Robotics: Robot Operating System (ROS), Gazebo, OpenAI Gym, simultaneous localization and mapping (SLAM), computer vision, motion planning, autonomous navigation

Tools: Linux, Git, Docker, Azure Databricks, Weights & Biases, Raspberry Pi